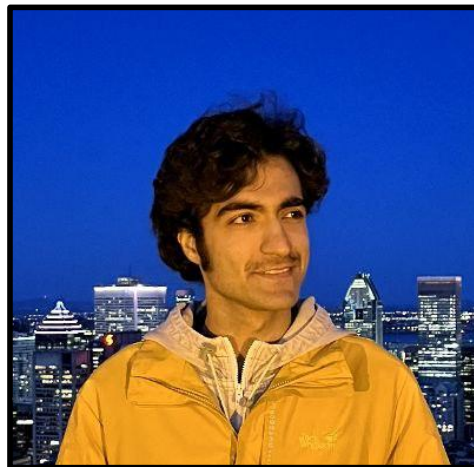
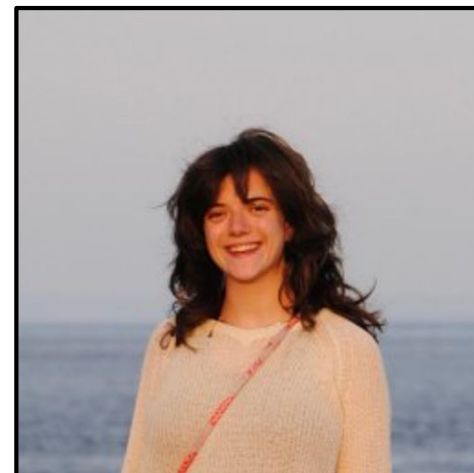




Uni-DAD: Unified Distillation and Adaptation of Diffusion Models for Few-step Few-shot Image Generation



Yara Bahram*
PhD student



Mélodie Desbos*
Master Student



Mohammadhadi Shateri
Professor



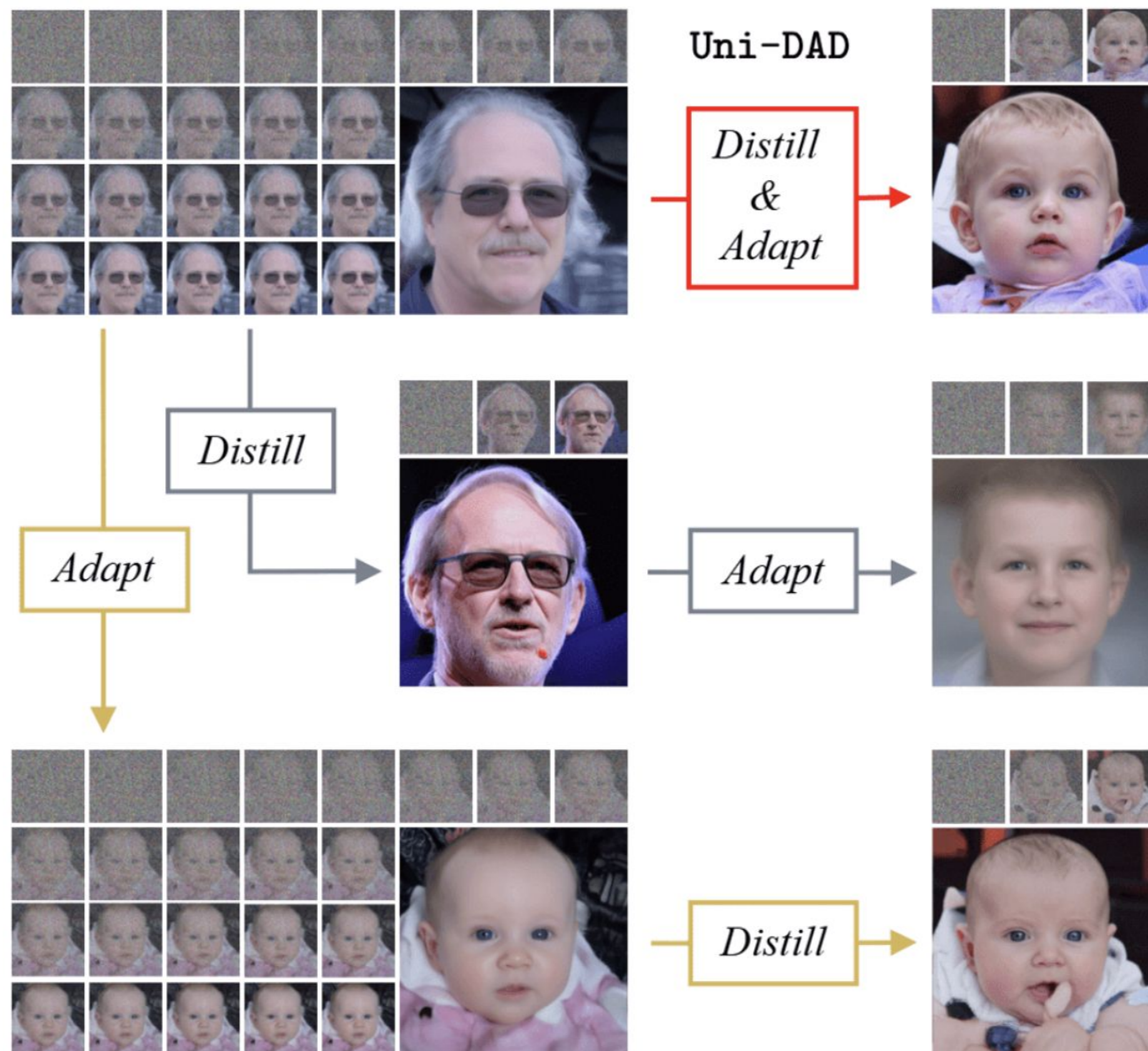
Eric Granger
Professor

*Equal Contribution

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Distillation and Adaptation of Diffusion Models



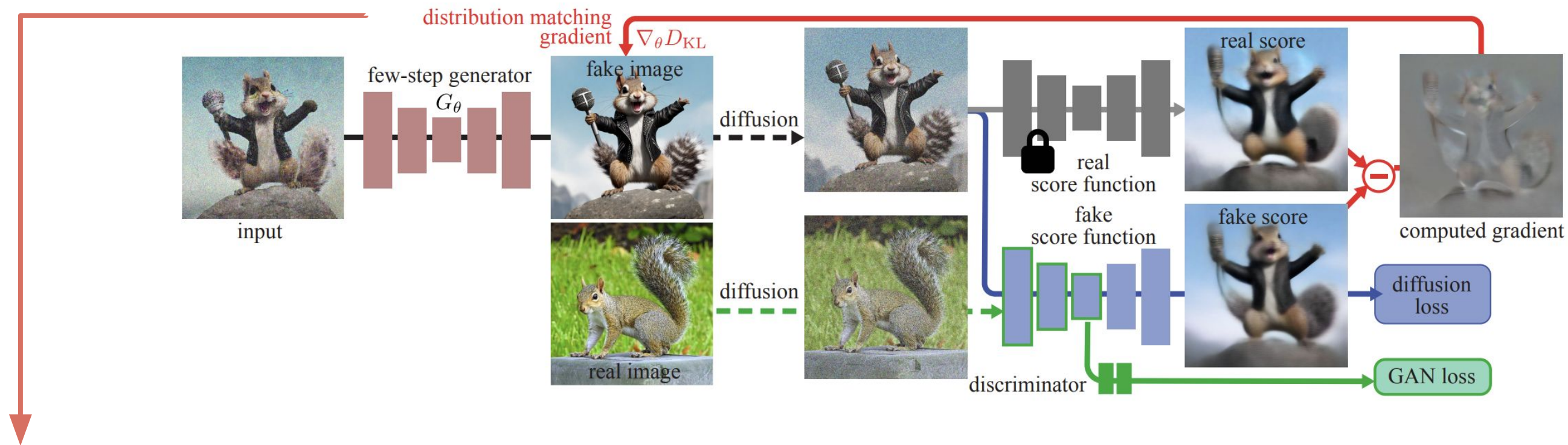
Need : Fast generation in new domains

Diffusion: adapts well but remains slow

- Two-stage pipelines:
 - Distill-then-Adapt → **Over-smoothing**
 - Adapt-then-Distill → **Over-fitting**

→ **Uni-DAD** : Unified Distillation and Adaptation
→ Fast generation
→ Diverse and high quality images

Background: Distribution Matching Distillation (DMD)



DMD loss guides towards the source teacher distribution

KL Divergence

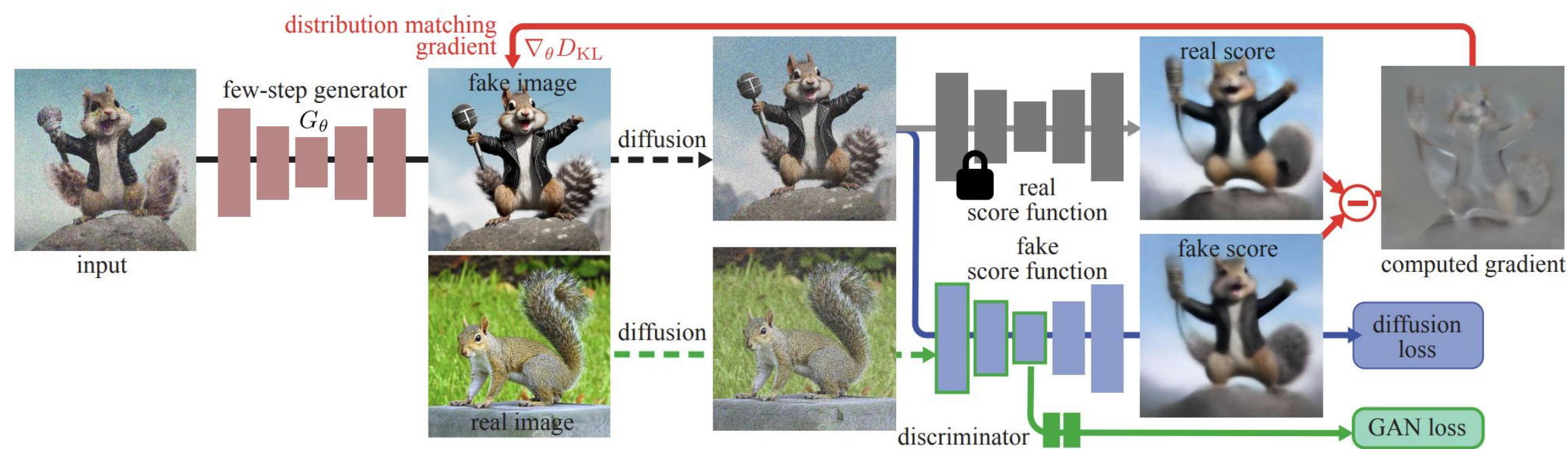
Student distribution

Student Generator

$$\nabla_{\theta} \mathcal{L}_{\text{DMD}} = \nabla_{\theta} D_{\text{KL}}(p^{\text{fk}} \| p^{\text{src}}) = \mathbb{E}_z \left[\underbrace{(\nabla_x \log p^{\text{fk}}(x) - \nabla_x \log p^{\text{src}}(x))}_{\text{Score function}} \frac{dG_{\theta}}{d\theta} \right]$$

Source distribution

Background: Distribution Matching Distillation (DMD)



GAN loss to distinguish target domain images from student generations

Generator

$$\mathcal{L}_{\text{GAN}}^G(\theta) = -\mathbb{E}_{t,z} [\log(h_{\psi}(f_{\phi}(x_t)))]$$

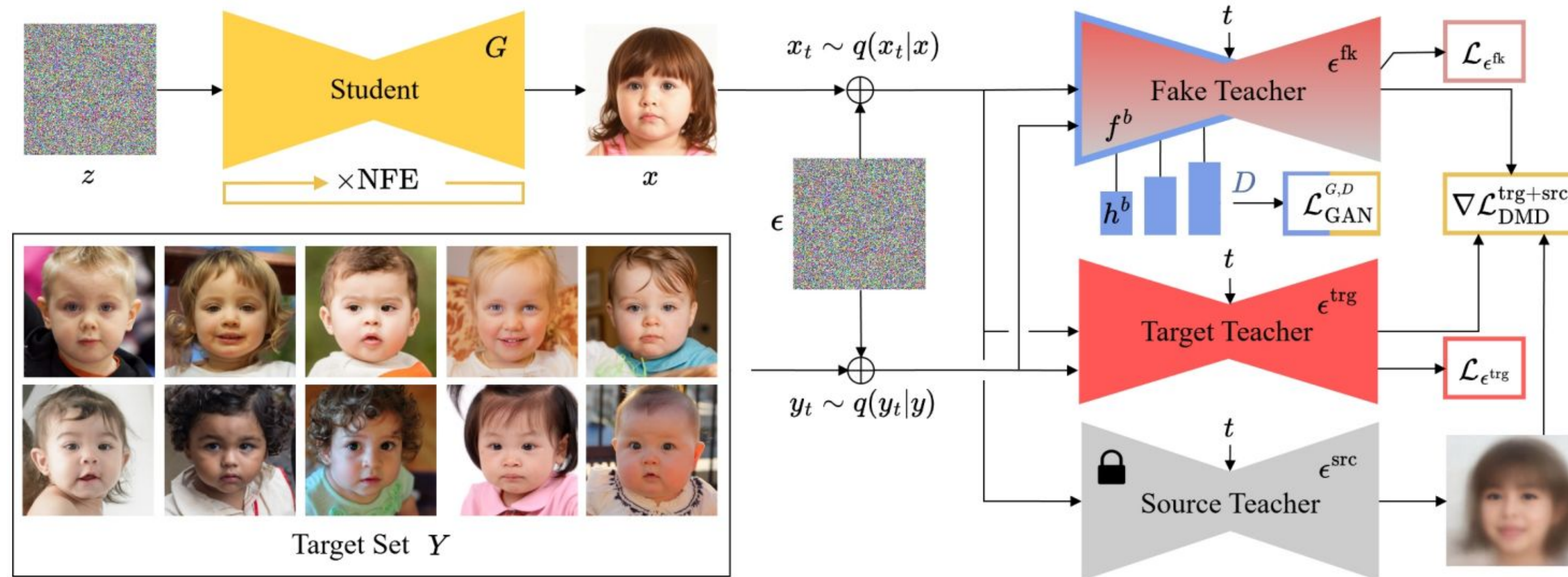
Classifier

Discriminator

$$\mathcal{L}_{\text{GAN}}^D(\psi, \phi) = -\mathbb{E}_{t,y} [\log(h_{\psi}(f_{\phi}(y_t)))] - \mathbb{E}_{t,z} [\log(1 - h_{\psi}(f_{\phi}(x_t)))]$$

Feature extractor

Uni-DAD Overview

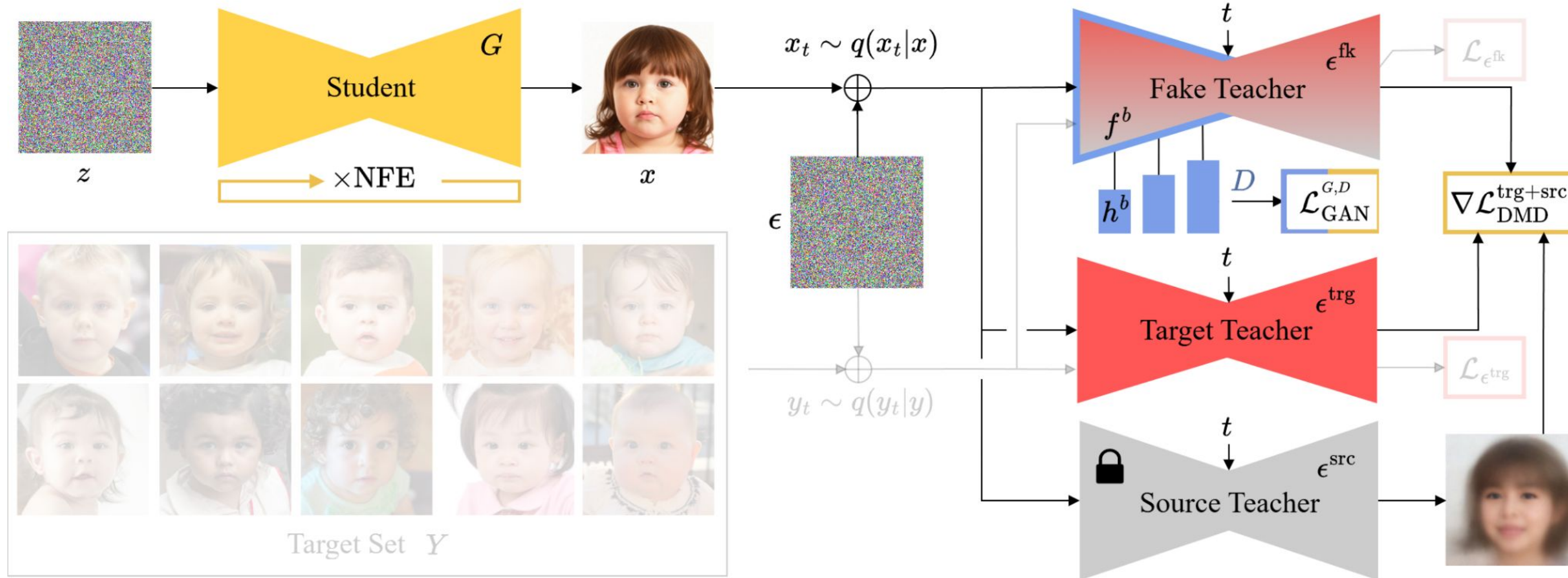


Student Generator for fast sampling on target domain

Dual-domain DMD for guidance towards the intersection of source and target domains

Multi-head GAN loss to distinguish target domain images from student generations

Uni-DAD Overview



(1) **Student** update:

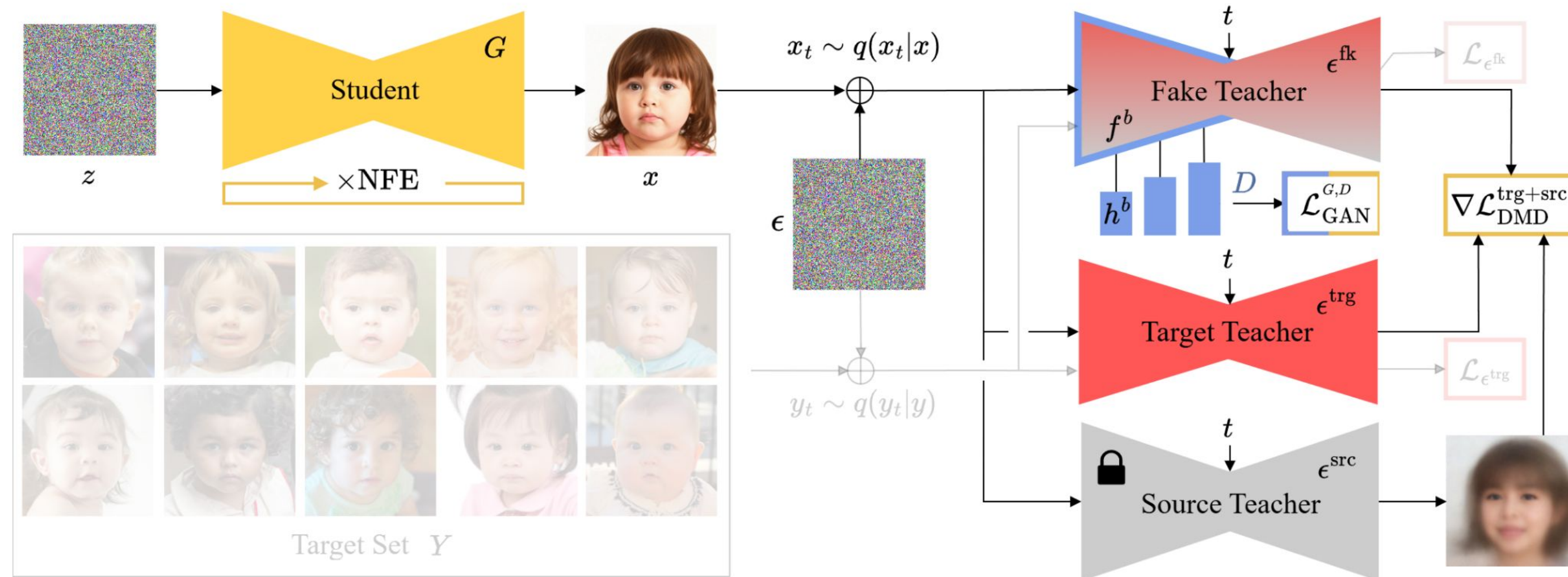
Source DMD:
$$\nabla_{\theta} \mathcal{L}_{\text{DMD}^{\text{src}}} \approx \mathbb{E}_{t,z} \left[\omega_t (\epsilon^{\text{fk}}(x_t) - \epsilon^{\text{src}}(x_t)) \frac{dG_{\theta}}{d\theta} \right]$$

Target DMD:
$$\nabla_{\theta} \mathcal{L}_{\text{DMD}^{\text{trg}}} \approx \mathbb{E}_{t,z} \left[\omega_t (\epsilon^{\text{fk}}(x_t) - \epsilon^{\text{trg}}(x_t)) \frac{dG_{\theta}}{d\theta} \right]$$

Dual-domain DMD:
$$\nabla_{\theta} \mathcal{L}_{\text{DMD}}^{\text{trg+src}} = (1 - a) \nabla_{\theta} \mathcal{L}_{\text{DMD}^{\text{src}}} + a \nabla_{\theta} \mathcal{L}_{\text{DMD}^{\text{trg}}}$$



Uni-DAD Overview



(1) **Student** update:

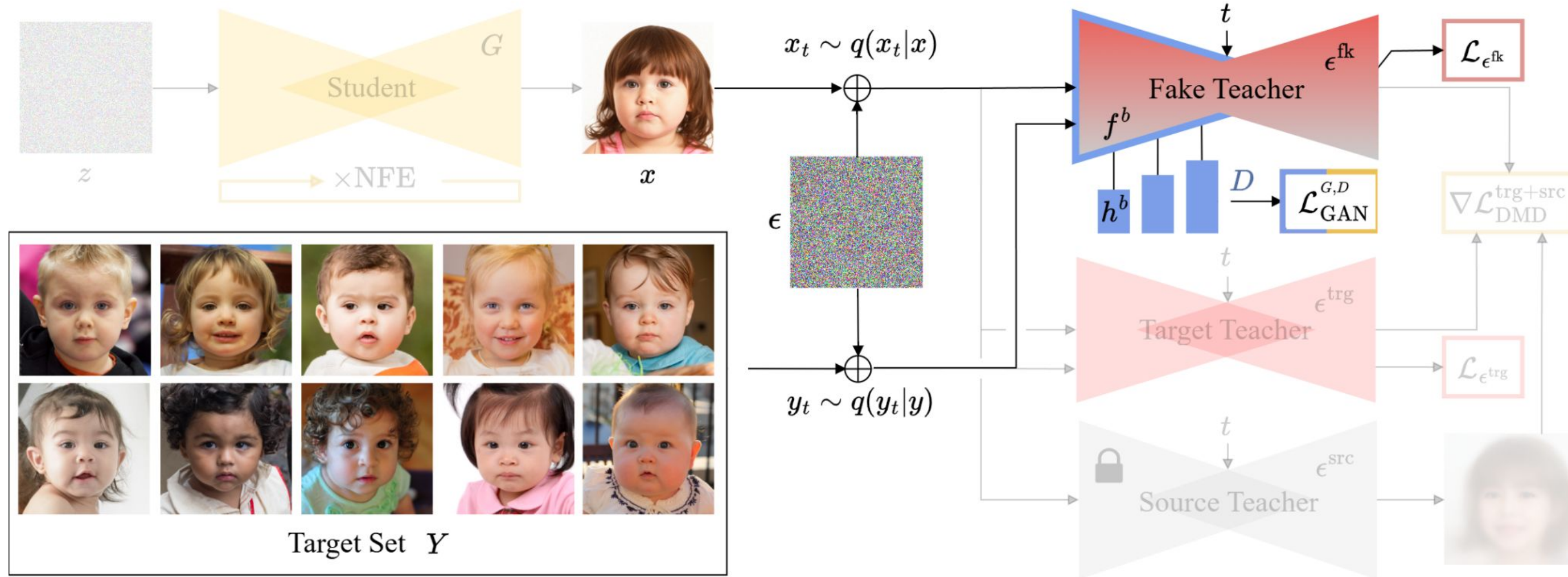
Multi-head GAN G: $\mathcal{L}_{\text{GAN}}^G(\theta) = -\mathbb{E}_{t,z} \sum_{b \in \mathcal{B}} \log(h_{\psi}^b(f_{\phi}^b(x_t)))$

Generator
Multi-head
Feature extractor
Linear classifier

G Loss: $\mathcal{L}_G(\theta) = \mathcal{L}_{\text{DMD}}^{\text{trg+src}}(\theta) + \lambda_{\text{GAN}}^G \mathcal{L}_{\text{GAN}}^G(\theta)$



Uni-DAD Overview



(2) **Discriminator** and **Fake Teacher** update:

Discriminator

Multi-head GAN D:

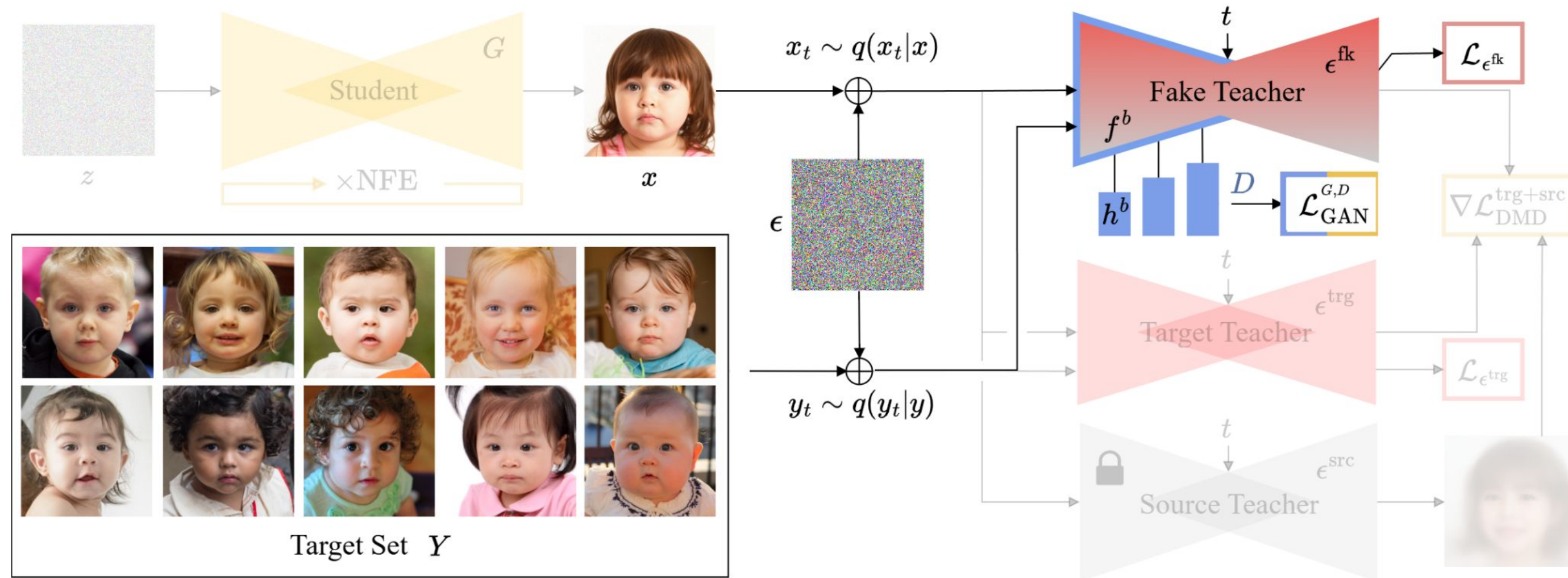
$$\mathcal{L}_{\text{GAN}}^D(\psi, \phi) = -\mathbb{E}_{t,y} \sum_{b \in \mathcal{B}} \log(h_{\psi}^b(f_{\phi}^b(y_t))) - \mathbb{E}_{t,z} \sum_{b \in \mathcal{B}} \log(1 - h_{\psi}^b(f_{\phi}^b(x_t)))$$

Fake Teacher:

$$\mathcal{L}_{\text{fk}}(\phi) = \mathbb{E}_{t,z} \left[\|\epsilon^{\text{fk}}_{\phi}(x_t) - \epsilon\|_2^2 \right]$$



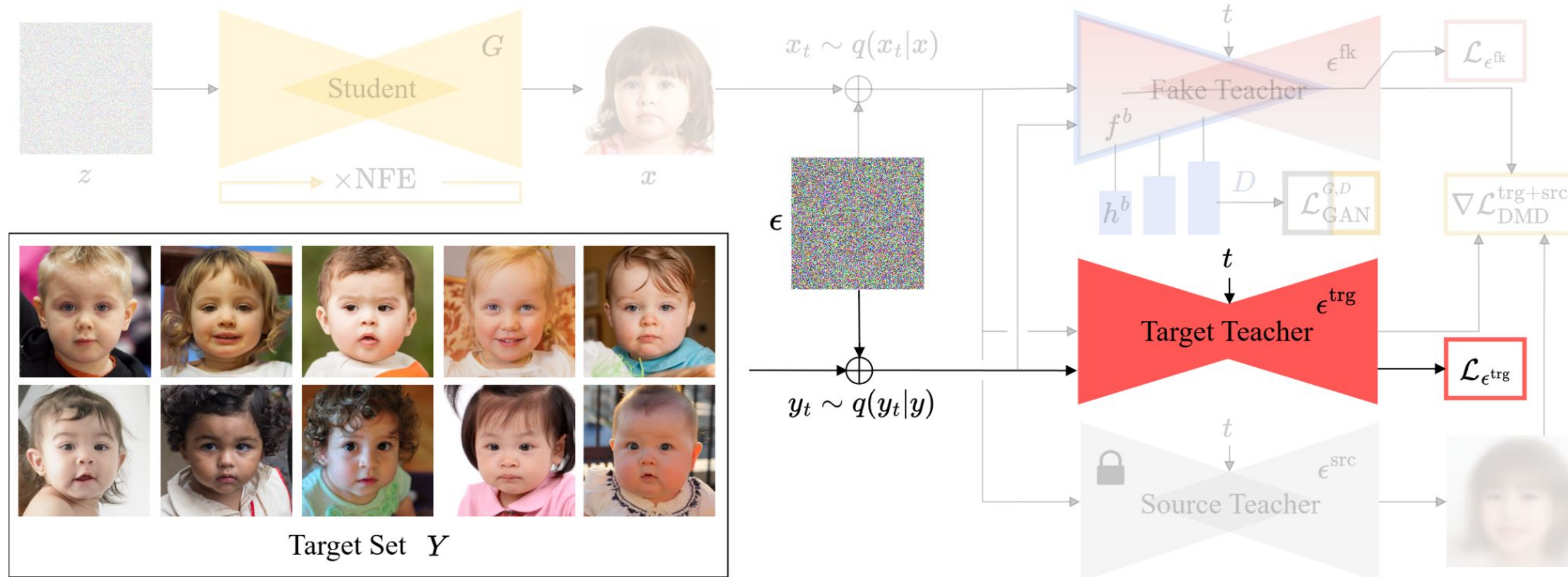
Uni-DAD Overview



(2) **Discriminator** and **Fake Teacher** update:

$$\text{fk+D Loss:} \quad \mathcal{L}_{\text{fk}+D}(\phi, \psi) = \mathcal{L}_{\text{fk}}(\phi) + \lambda_{\text{GAN}}^D \mathcal{L}_{\text{GAN}}^D(\psi, \phi)$$

Uni-DAD Overview



(3) (Optional) **Target Teacher** update:

trg Loss:

$$\mathcal{L}_{\text{trg}}(\eta) = \mathbb{E}_{t, \epsilon, y} \left[\left\| \epsilon_{\eta}^{\text{trg}}(y_t) - \epsilon \right\|_2^2 \right]$$



Applications

(i) Unconditional Generation for 10 shot adaptation

Babies	Close Domains
	Target Images
.....	
	DDPM-PA
	CRDI
	FT
	DMD2-FT
	FT-DMD2
	Uni-DAD (no ϵ^{trg})
	Uni-DAD

(ii) Subject Driven Personalization for 4-6 shot adaptation

Target Instances	Uni-DAD SDv1.5, NFE = 1	DreamBooth SDv1.5, NFE = 2 × 50	DMD2-DreamBooth SDv1.5, NFE = 1	PSO SDXL, NFE = 4
	 “a prt dog with a mountain in the background”			
	 “a prt dog in the snow”			
“a prt dog”	 “a prt dog with a blue house in the background”			



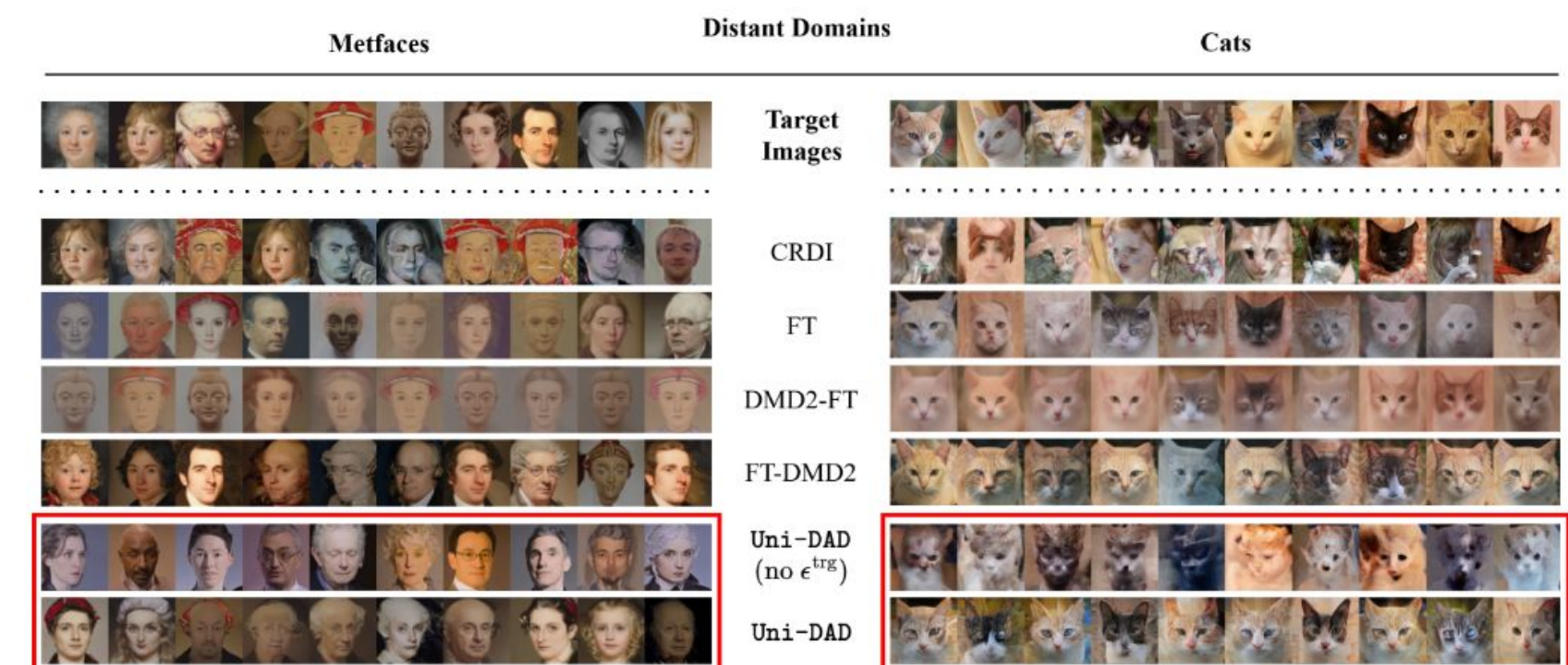
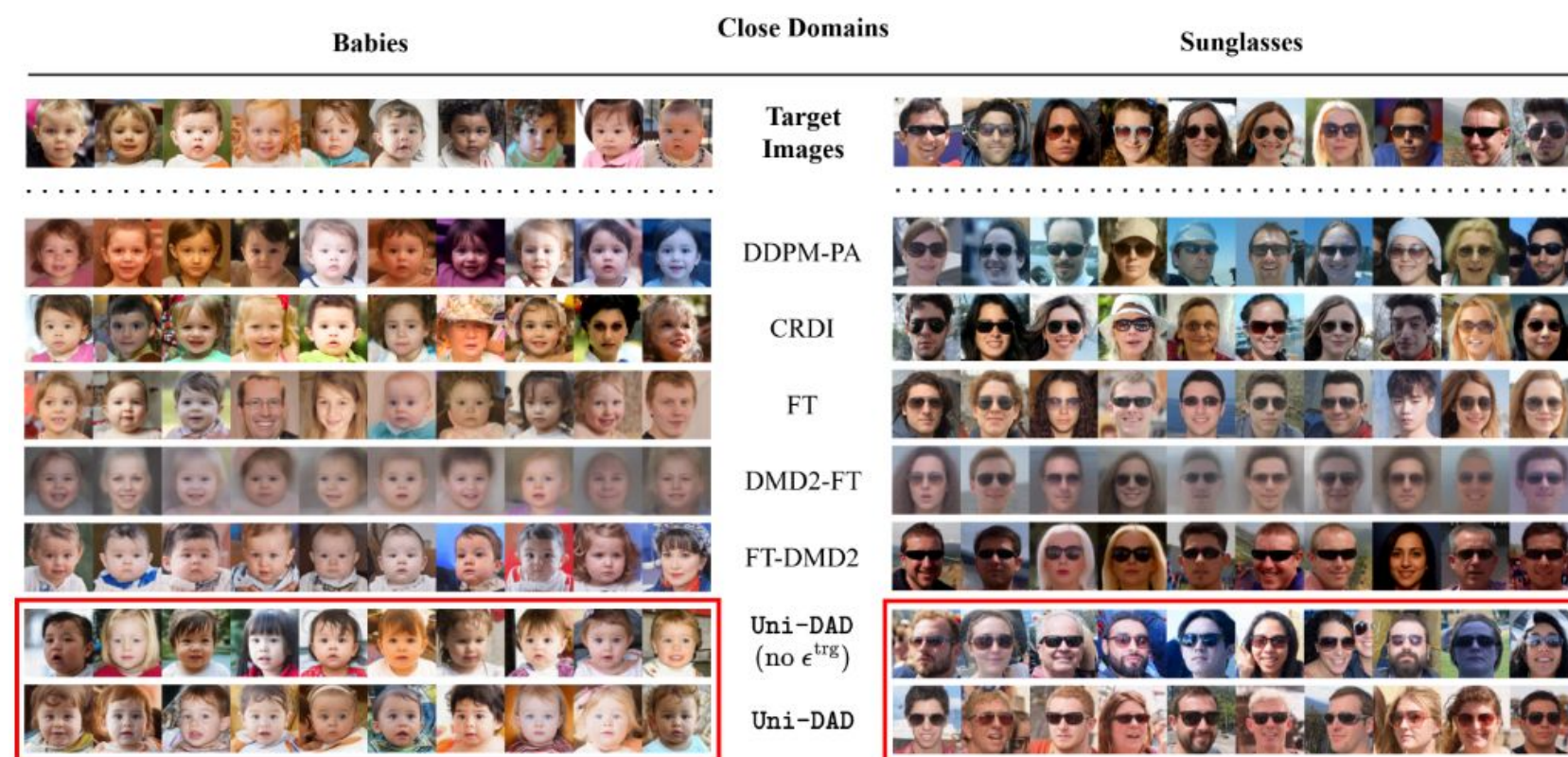
Applications - Unconditional Generation

- Using **Guided DDPM**
- Pretrained on **FFHQ Dataset** (70K diverse faces)
- Tested on **10 preselected samples** of 4 target domains
- Evaluated with **FID score** and **Intra-LPIPS**

→ Better or comparable quality than SoTA

→ ≤ 4 denoising steps

→ Often surpasses two-stage methods in **quality** and **diversity**



Applications - Unconditional Generation

- Using **Guided DDPM** → Better or comparable quality than SoTA
- Pretrained on **FFHQ Dataset** (70K diverse faces) → ≤ 4 denoising steps
- Tested on **10 preselected samples** of 4 target domains → Often surpasses two-stage in **quality** and **diversity**
- Evaluated with **FID score** and **Intra-LPIPS**

Variant	Method	NFE↓	End-to-end	FID ↓				Intra-LPIPS ↑			
				Babies	Sunglasses	MetFaces	Cats	Babies	Sunglasses	MetFaces	Cats
<i>Non-distilled</i>	DDPM-PA [52]	1000	✓	48.92	34.75	—	—	<u>0.59</u>	<u>0.60</u>	—	—
	CRDI [3]	25	✓	48.52	24.62	121.36 [†]	220.95	0.52	0.50	0.41	<u>0.51</u>
	FT	25	✓	57.06	37.86	72.99	61.62	0.32	0.48	<u>0.45</u>	0.42
<i>Distilled</i>	DMD2-FT	3		140.27	77.49	129.26	89.32	0.08	0.20	0.08	0.18
	FT-DMD2	3		57.11	41.85	63.25	51.85	0.42	0.42	0.44	0.34
	Uni-DAD (no ϵ^{trg})	3	✓	47.38	22.57	72.18	199.91	0.45	0.51	0.42	0.40
	Uni-DAD	3	✓	45.09	24.45	58.13	55.32	0.46	0.54	0.36	0.36

[†] We were unable to reproduce CRDI's reported FID of 94.86 [3]. However, our qualitative samples and Intra-LPIPS closely match their findings.



Applications - Unconditional Generation

- Using **Guided DDPM**
 - **Significantly lower generation cost** than *non-distilled*
- Pretrained on **FFHQ Dataset** (70K diverse faces)
 - **Lower training cost** than *two-stage* baselines
- Tested on **10 preselected samples** of 4 target domains
 - Parameter-efficiency is future work
- Evaluated with **Training and Test Cost**

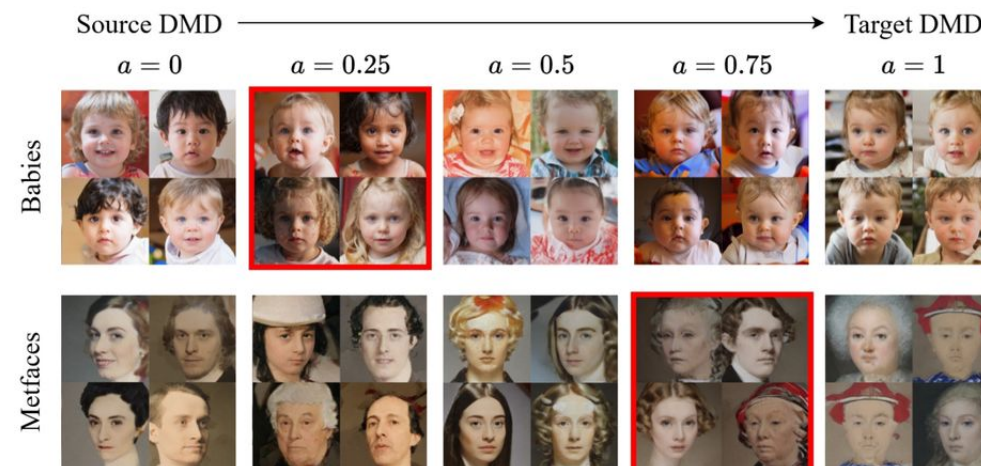
Variant	Method	Training Cost (20K iters, †: not applicable)			Test Cost (5K generations, batch size 10)			
		GPU · h	GPU	Mem [GB]	Time [m]	Mem [GB]	TFLOPs/img	NFE
<i>Non-distilled</i>	CRDI †	1	4 A100	124.7	63	14	55.7	25
	FT	0.8	1 H100	27.6	35	14	55.7	25
<i>Distilled</i>	FT-DMD2	3	1 H100	40.4	4.2	13	2.2	3
	DMD2-FT	3	1 H100	40.4	4.2	13	2.2	3
	Uni-DAD (No ϵ^{trg})	2.2	1 H100	40.3	4.2	13	2.2	3
	Uni-DAD	2.8	1 H100	48.8	4.2	13	2.2	3

Ablations

Sample quality to NFE variations



Dual Domain DMD weight factor



Available checkpoints at initialization

Available Checkpoints		FID ↓	
<i>Pre-distilled G</i>	<i>Pre-adapted ϵ^{trg}</i>	Babies	MetFaces
		45.09	58.13
✓		42.68	77.80
	✓	46.73	62.67
✓	✓	42.04	54.01

Ablations of Uni-DAD components

Group	Components				FID ↓	
	DMD ^{src}	DMD ^{trg}	GAN ^{Mh}	GAN ^{Sh}	B	M
<i>GAN_only</i>				✓	56.90	80.14
			✓		130.34	110.00
<i>DMD_only</i>		✓			110.39	68.05
	✓				166.99	105.67
	✓	✓			84.19	69.80
<i>DMD+GAN</i>		✓		✓	54.18	68.54
	✓			✓	57.58	83.67
	✓	✓		✓	48.89	80.24
		✓	✓		54.68	68.74
	✓		✓		47.38	64.13
	✓	✓	✓		45.09	58.13

Target size and NFE

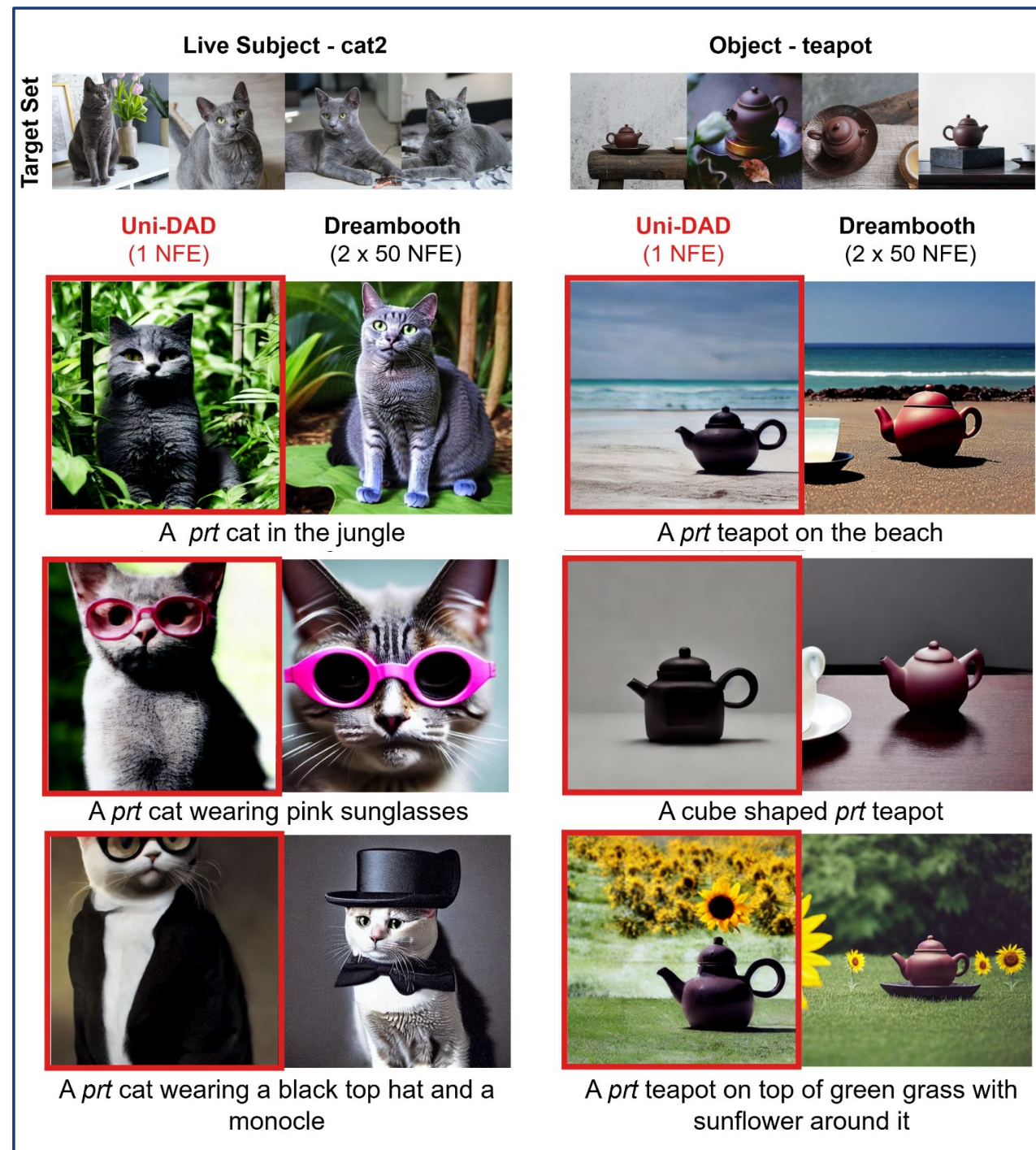
Methods	NFE↓	1-shot		5-shot		10-shot	
		B	M	B	M	B	M
CRDI	25	105.51	145.10	51.71	126.34	48.52	121.36
Uni-DAD	4	72.38	95.44	45.86	81.85	41.39	59.49
Uni-DAD	3	90.33	90.29	52.73	83.69	45.09	58.13
Uni-DAD	2	95.69	114.78	68.75	98.36	62.45	79.08
Uni-DAD	1	109.55	132.79	93.52	103.84	98.52	89.03

GAN losses evaluation

GAN Loss	Multi-head		Single-head	
	Babies	MetFaces	Babies	MetFaces
BCE	45.09	58.13	48.89	80.24
Hinge	45.82	64.83	50.92	72.11
LSGAN	46.18	91.55	47.98	64.36
WGAN	64.54	95.35	58.50	84.91



Applications - Subject Driven Personalization



- **Stable Diffusion v1.5**
- Pretrained on **LAION-5B Dataset** (paired text-image)
- Tested with **Dreambooth Benchmark**
- Evaluated with **CLIP-I/T, DINO, Intra-LPIPS, Inter-LPIPS**

Applications - Subject Driven Personalization

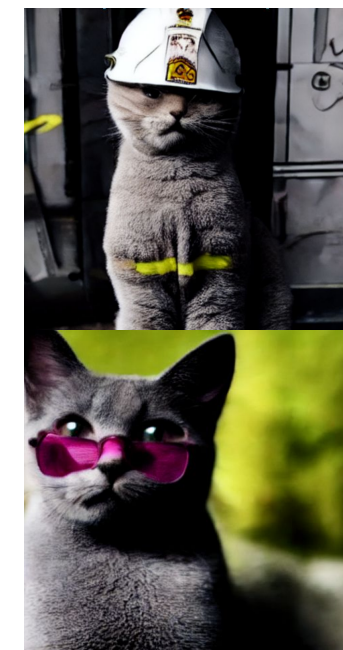
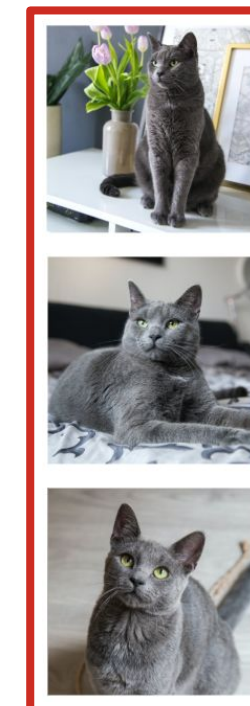
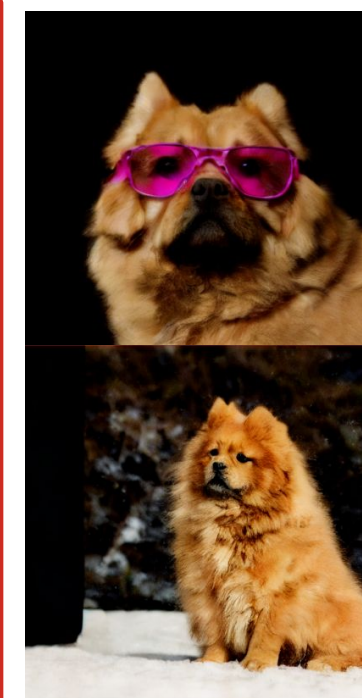
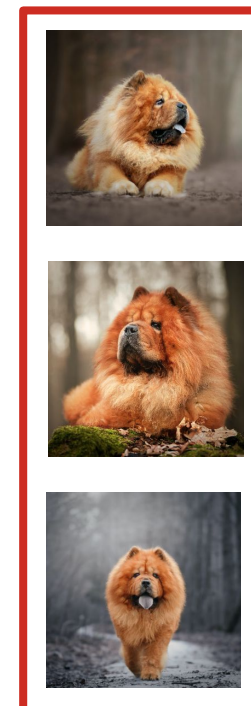
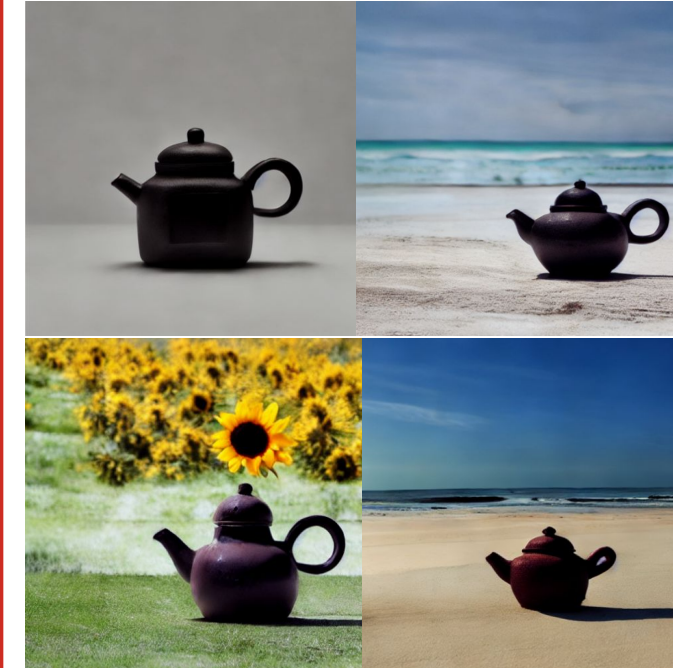
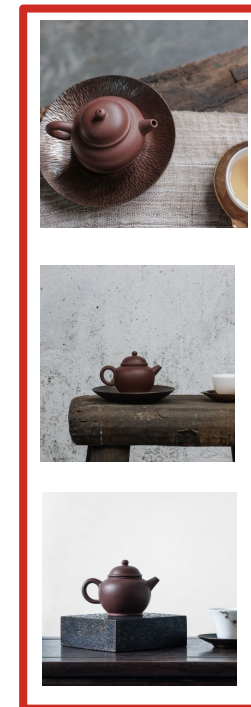
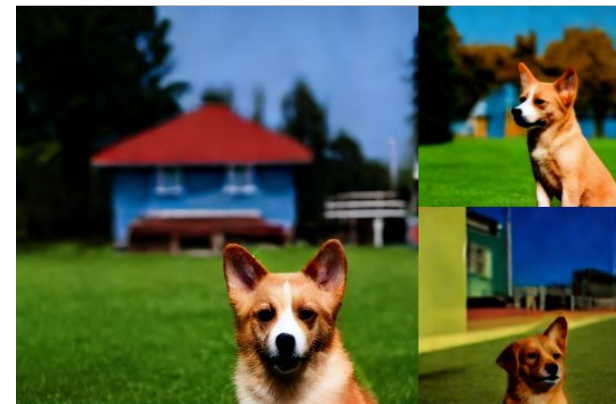
- 1 denoising step
- Competes with *non-distilled* DreamBooth
- **Best trade-off** among distilled baselines

Variant	Method	NFE↓	1-stage	Quality			Diversity	
				DINO↑	CLIP-I↑	CLIP-T↑	Intra-LPIPS↑	Inter-LPIPS↑
<i>Non-distilled</i>	FT [34]	2×50	✓	0.58	0.77	0.32	0.67 ± 0.08	0.73 ± 0.06
	Turbo-PSO _{SDXL} [27]	4		0.50	0.70	0.30	0.42 ± 0.07	0.60 ± 0.08
<i>Distilled</i>	DMD2-PSO _{SDv1.5} [27]	1		0.14	0.56	0.23	0.07 ± 0.02	0.11 ± 0.02
	DMD2-FT	1		0.20	0.61	<u>0.26</u>	0.58 ± 0.07	0.70 ± 0.09
	FT-DMD2	1		0.57	0.75	0.25	0.22 ± 0.04	0.25 ± 0.07
	Uni-DAD	1	✓	<u>0.47</u>	<u>0.73</u>	0.29	<u>0.51 ± 0.09</u>	<u>0.59 ± 0.09</u>



Applications - Subject Driven Personalization

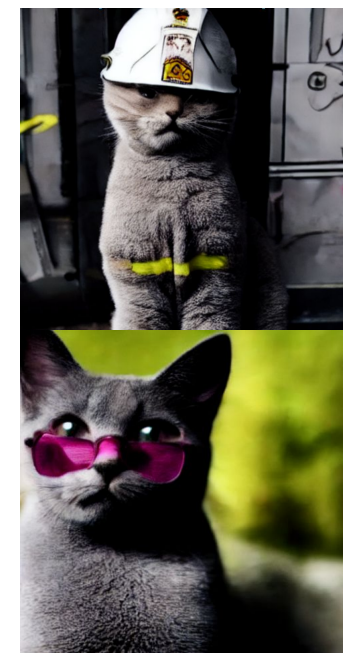
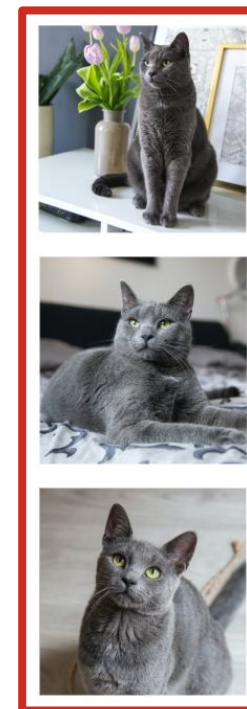
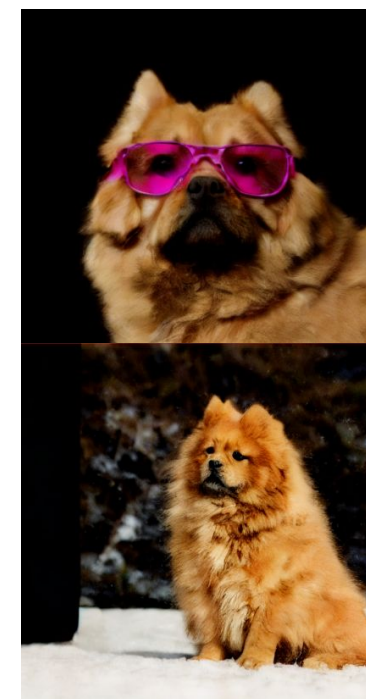
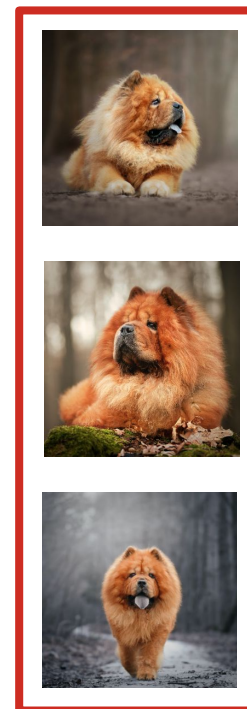
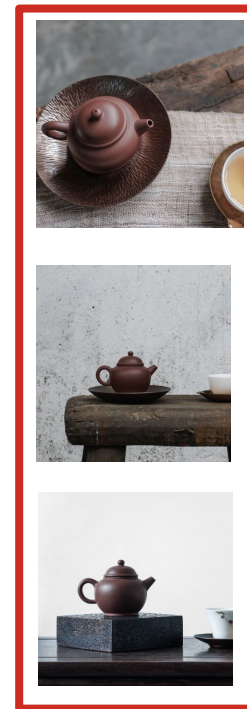
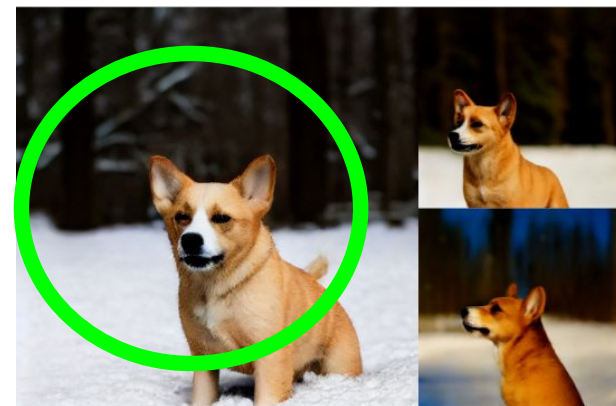
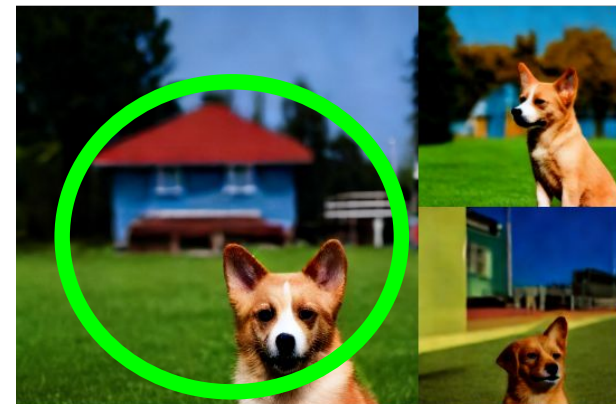
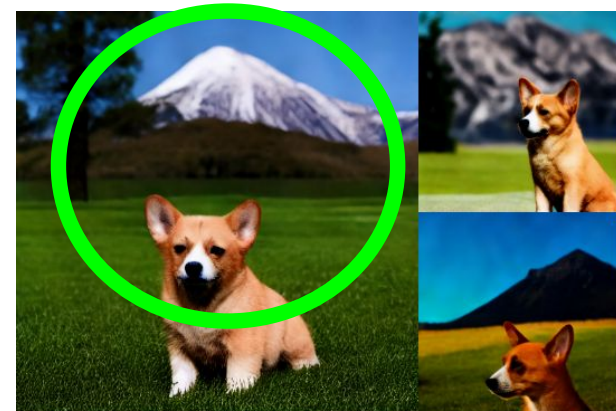
→ **Best trade-off** between **subject fidelity**, **prompt alignment**, and **diversity**



- Recontextualization
- Accessorization
- Property Modification

Applications - Subject Driven Personalization

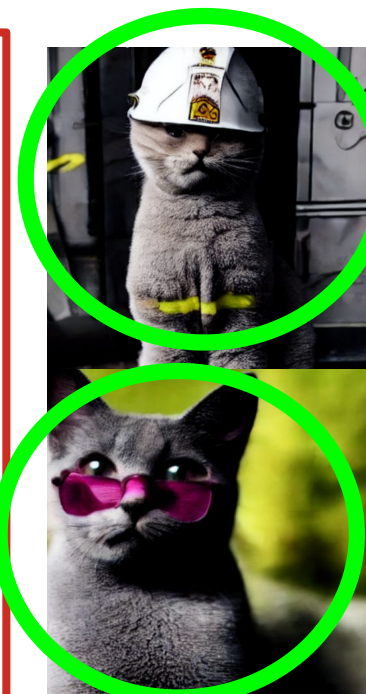
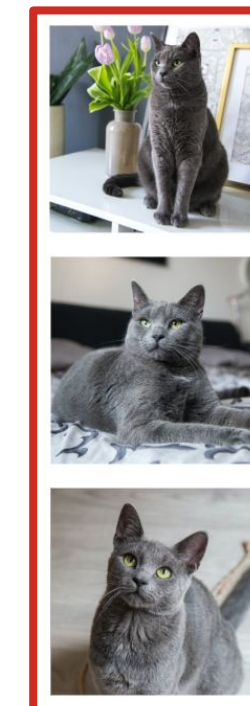
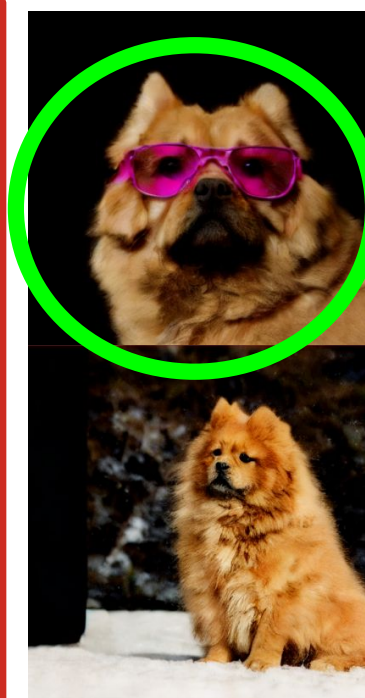
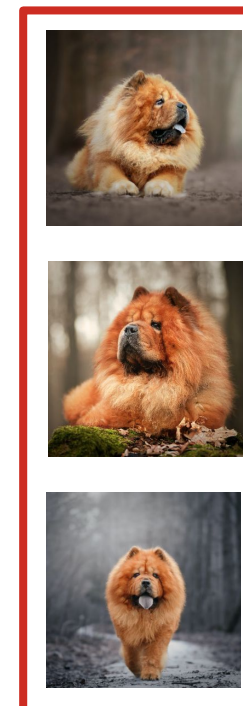
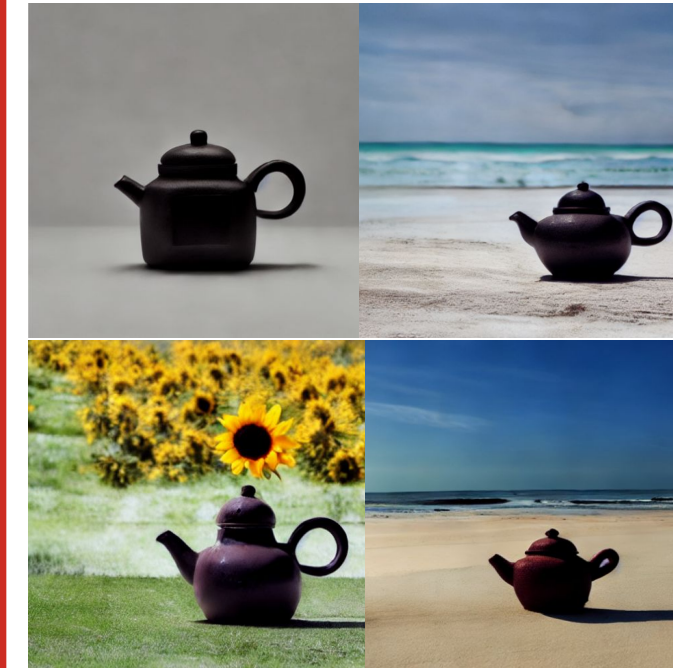
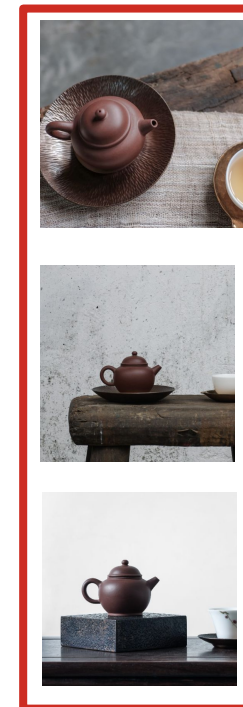
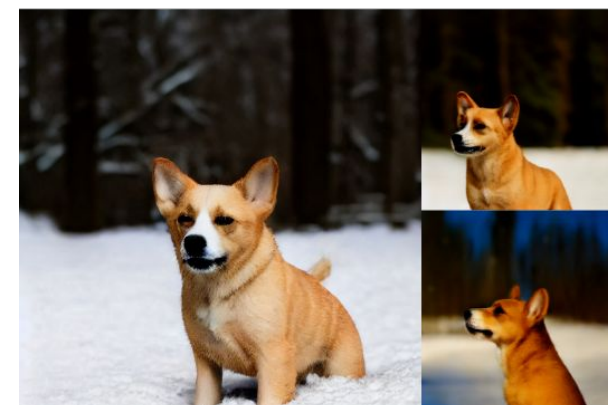
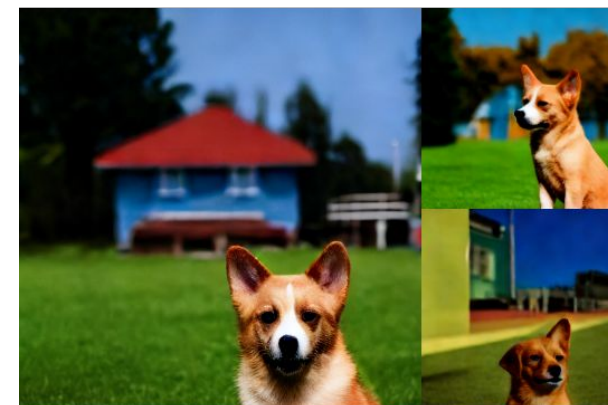
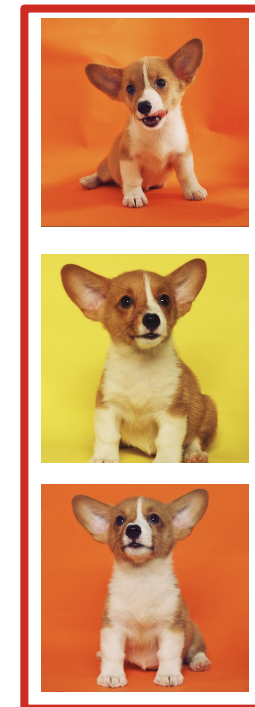
→ Best trade-off between subject fidelity, prompt alignment, and diversity



- Recontextualization
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- Property Modification

Applications - Subject Driven Personalization

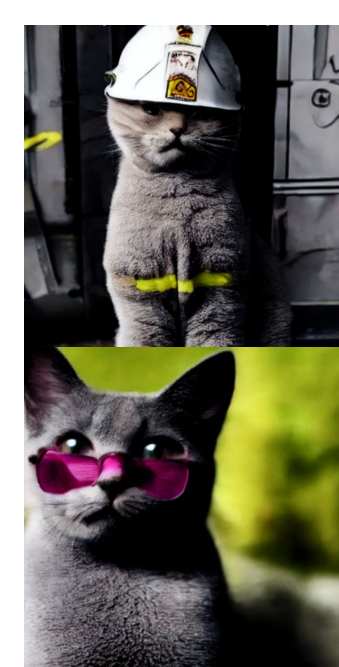
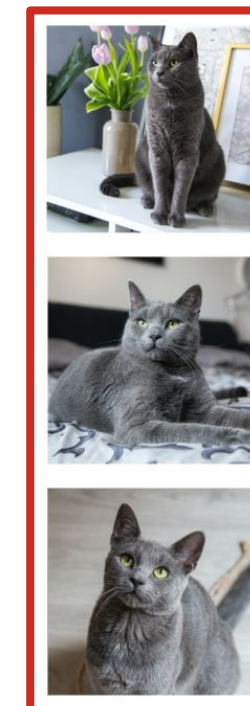
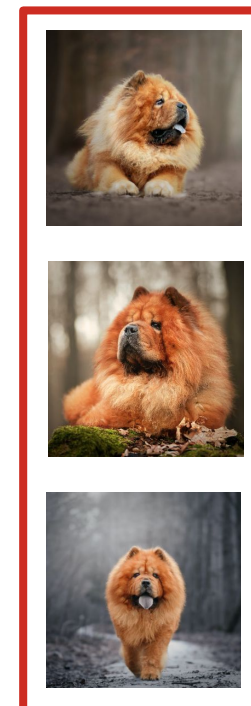
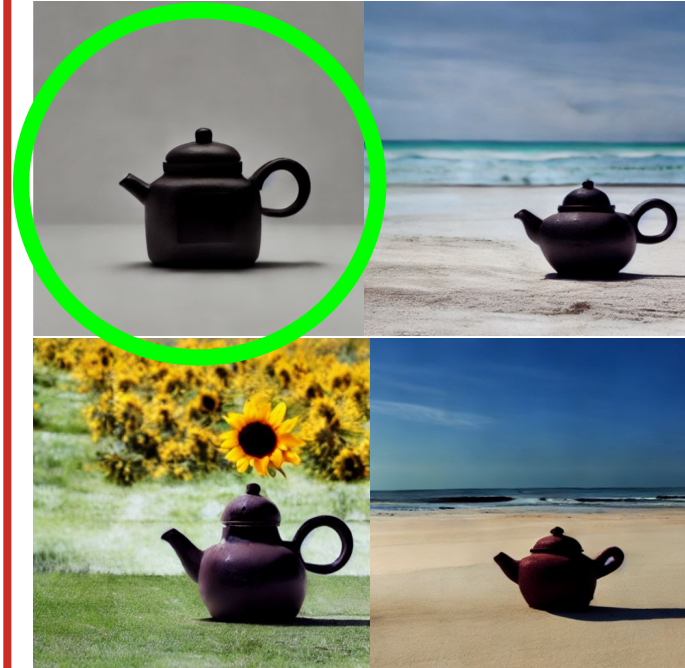
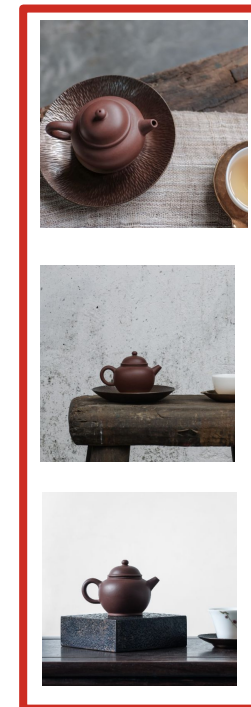
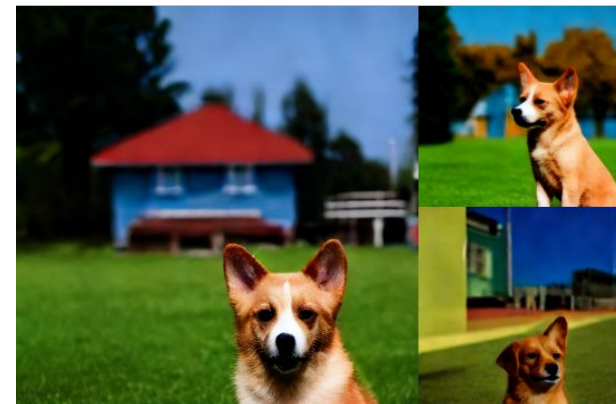
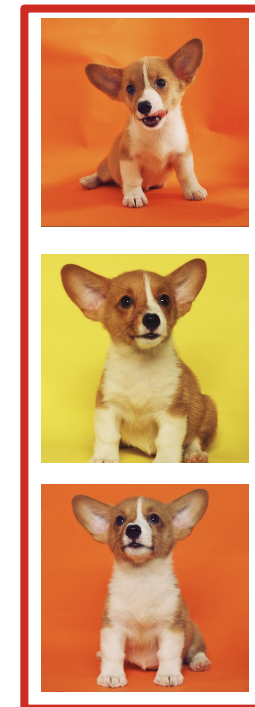
→ Best trade-off between subject fidelity, prompt alignment, and diversity



- Recontextualization
- Accessorization
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Applications - Subject Driven Personalization

→ Best trade-off between subject fidelity, prompt alignment, and diversity



- Recontextualization
- Accessorization
- Property Modification

Applications - Subject Driven Personalization

→ **Best trade-off** between subject fidelity, prompt alignment, and **diversity**



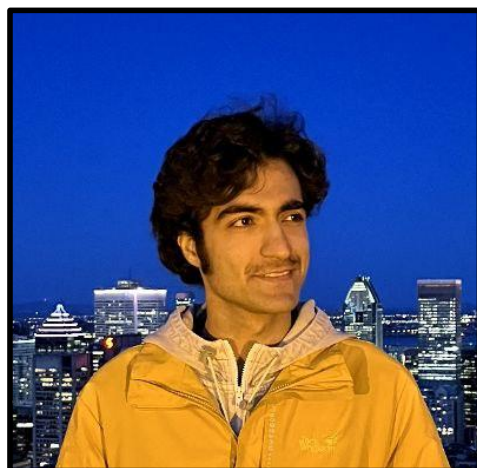
Uni-DAD: Unified **Distillation and Adaptation of Diffusion Models** for **Few-step Few-shot** Image Generation



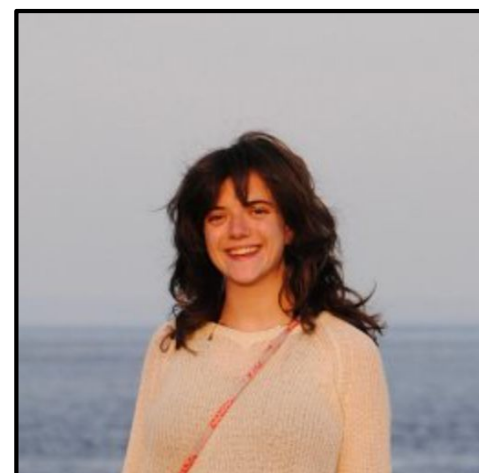
Contact us : yara.mohammadi-bahram@livia.etsmtl.ca
melodie.desbos@livia.etsmtl.ca



Uni-DAD: Unified **Distillation and Adaptation** of Diffusion Models for **Few-step Few-shot** Image Generation



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Mélodie Desbos*
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Mohammadhadi Shateri
Professor



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Professor

***Equal Contribution**

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Training Algorithm

Supplementary

Algorithm 1: Uni-DAD Training Iteration

Input : Source teacher ϵ^{src} , Optional target teacher ϵ^{trg} , Target set $Y = \{y\}$, Weight factor a , Training *step*, Update *ratio*

Output: Student G (Adapted and Distilled)

```
1  $\epsilon^{\text{fk}} \leftarrow \epsilon^{\text{src}}$ 
2 if  $\epsilon^{\text{trg}} == \emptyset$  then
3    $\epsilon^{\text{trg}} \leftarrow \epsilon^{\text{src}}$ ;  $\text{train\_target} \leftarrow \text{True}$ 

  // Prepare data
4  $t \sim \mathcal{U}\{0.02T, 0.98T\}$ 
5  $(z, \epsilon) \sim \mathcal{N}(0, I)$ 
6  $y_t \leftarrow q(y_t | y), y \sim Y$  // real
7  $x_t \leftarrow q(x_t | x), x \leftarrow G(z)$  // fake

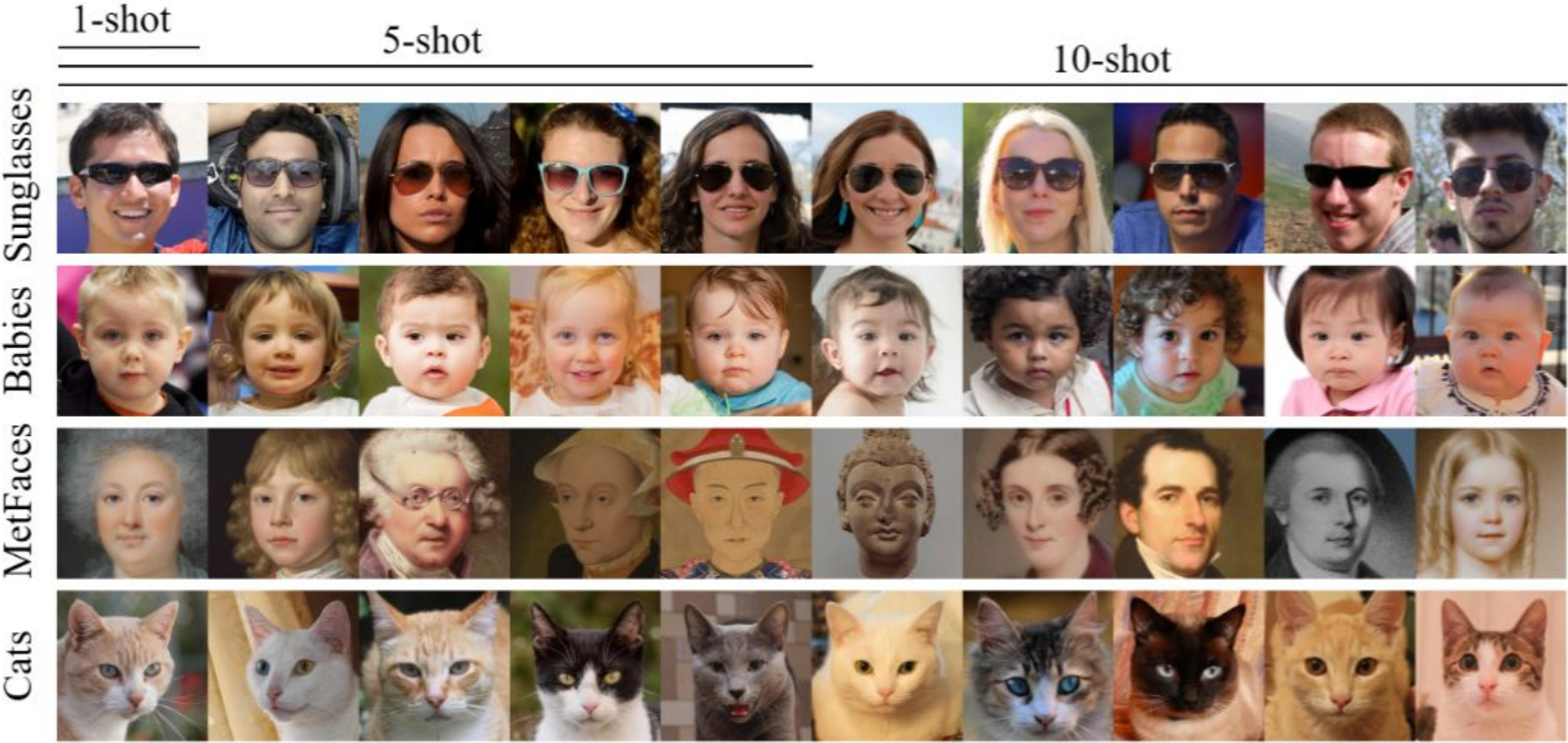
  // Student
8 if  $\text{step \% ratio} == 0$  then
9    $\mathcal{L}_{\text{DMD}}^{\text{trg+src}} \leftarrow \text{DualDMD}(x_t, \epsilon, a)$  // Eq 5
10   $\mathcal{L}_{\text{GAN}}^G \leftarrow \text{MhGAN}(x_t)$  // Eq 8
11   $\mathcal{L}_G \leftarrow \mathcal{L}_{\text{DMD}}^{\text{trg+src}} + \lambda_{\text{GAN}}^G \mathcal{L}_{\text{GAN}}^G$  // Eq 10
12   $G \leftarrow \text{update}(G, \mathcal{L}_G)$ 

  // Fake Teacher & Discriminator
13  $\mathcal{L}_{\text{fk}} \leftarrow \text{MSE}(\epsilon^{\text{fk}}(\text{stop\_grad}(x_t)), \epsilon)$  // Eq 6
14  $\mathcal{L}_{\text{GAN}}^D \leftarrow \text{MhGAN}(\text{stop\_grad}(x_t), y_t)$  // Eq 9
15  $\mathcal{L}_{\text{fk}+D} \leftarrow \mathcal{L}_{\text{fk}} + \lambda_{\text{GAN}}^D \mathcal{L}_{\text{GAN}}^D$  // Eq 11
16  $\epsilon^{\text{fk}} \leftarrow \text{update}(\epsilon^{\text{fk}}, \mathcal{L}_{\text{fk}+D})$ 

  // Target Teacher
17 if  $\text{step \% ratio} == 0$  &  $\text{train\_target}$  then
18    $\mathcal{L}_{\epsilon^{\text{trg}}} \leftarrow \text{MSE}(\epsilon^{\text{trg}}(y_t), \epsilon)$  // Eq 7
19    $\epsilon^{\text{trg}} \leftarrow \text{update}(\epsilon^{\text{trg}}, \mathcal{L}_{\epsilon^{\text{trg}}})$ 
```

FSIG Target Sets

Supplementary



Uni-DAD in Style Transfer

Supplementary

Style

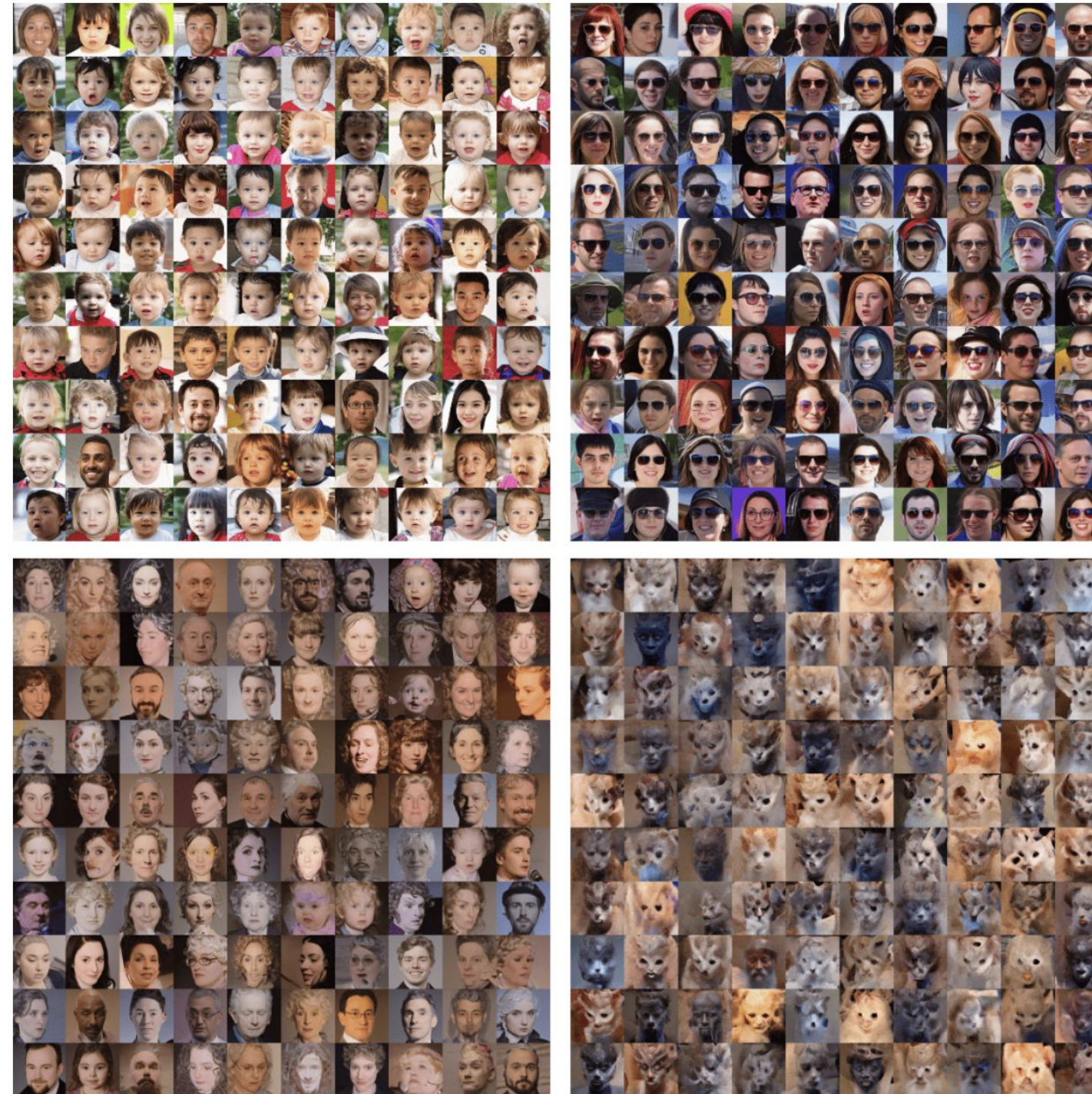


Generated



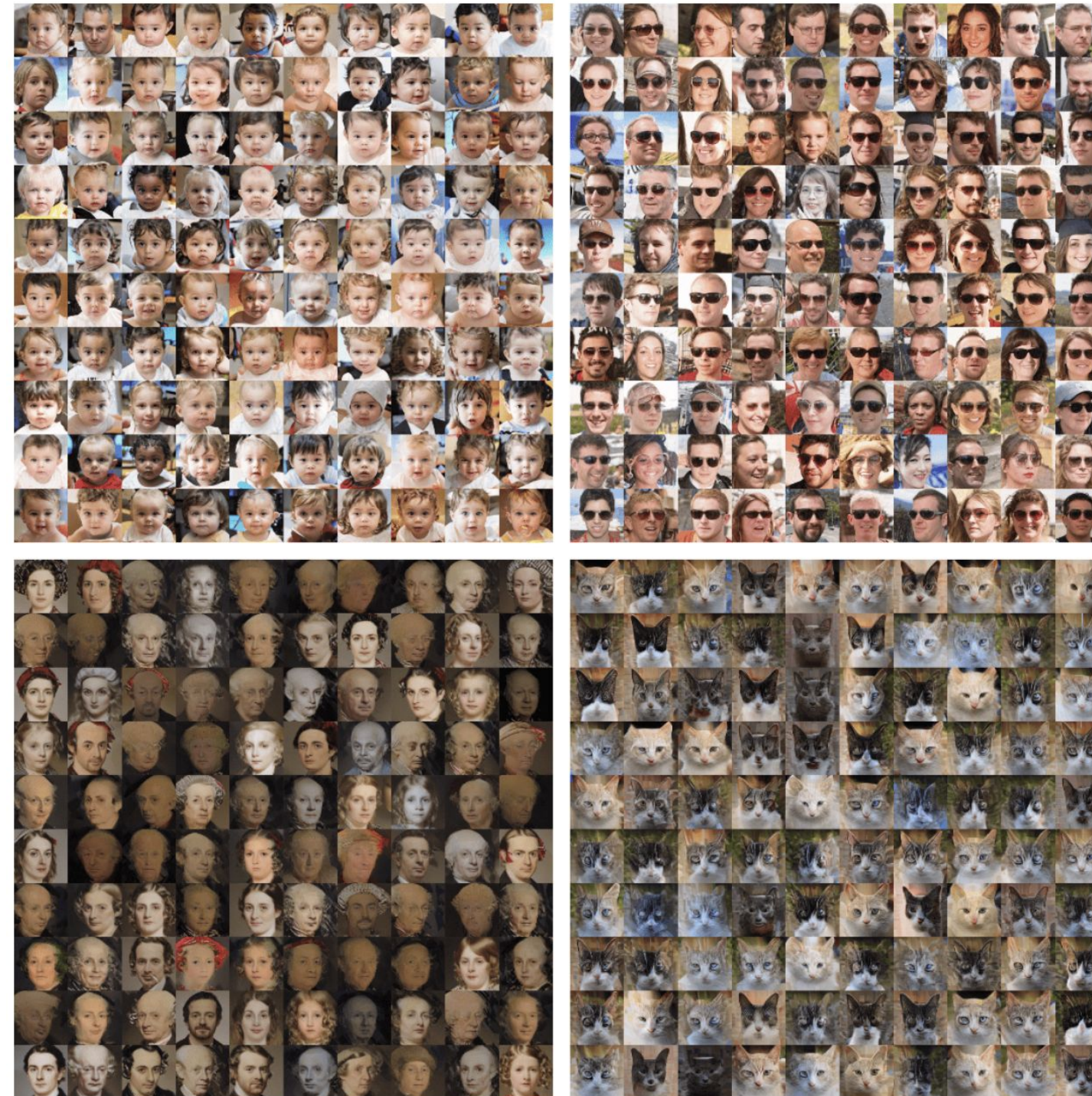
FSIG Generations without the Target Teacher

Supplementary



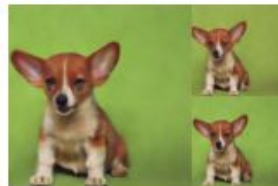
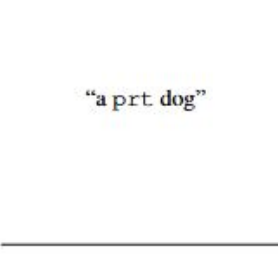
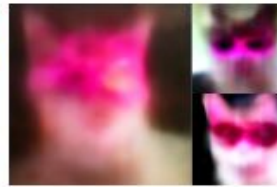
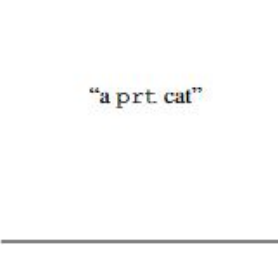
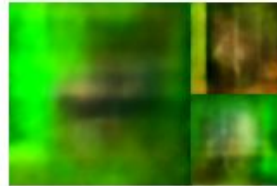
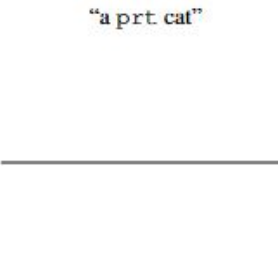
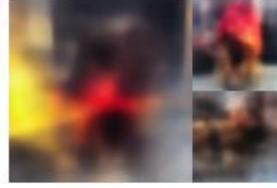
FSIG Generations with the Target Teacher

Supplementary



Qualitative Comparison in SDP

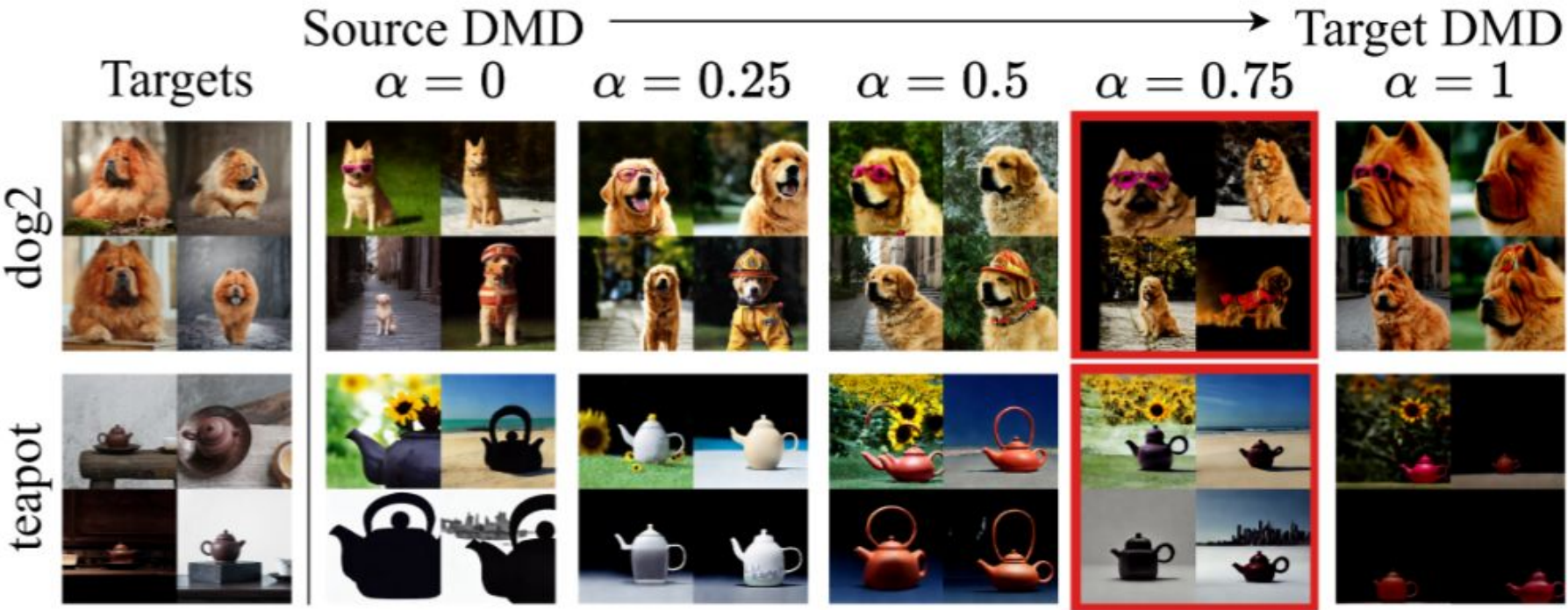
Supplementary

Target Instances	Uni-DAD SDv1.5, NFE = 1	DreamBooth SDv1.5, NFE = 2×50	DMD2-DreamBooth SDv1.5, NFE = 1	DreamBooth-DMD2 SDv1.5, NFE = 1	PSO SDXL, NFE = 4
					
	“a prt dog with a mountain in the background”				
					
	“a prt dog in the snow”				
					
	“a prt dog with a blue house in the background”				
					
	“a prt cat wearing pink glasses”				
					
	“a prt cat in the jungle”				
					
	“a prt cat in a firefighter outfit”				



Ablating DMD weight in SDP

Supplementary



One-step Generation with DMD

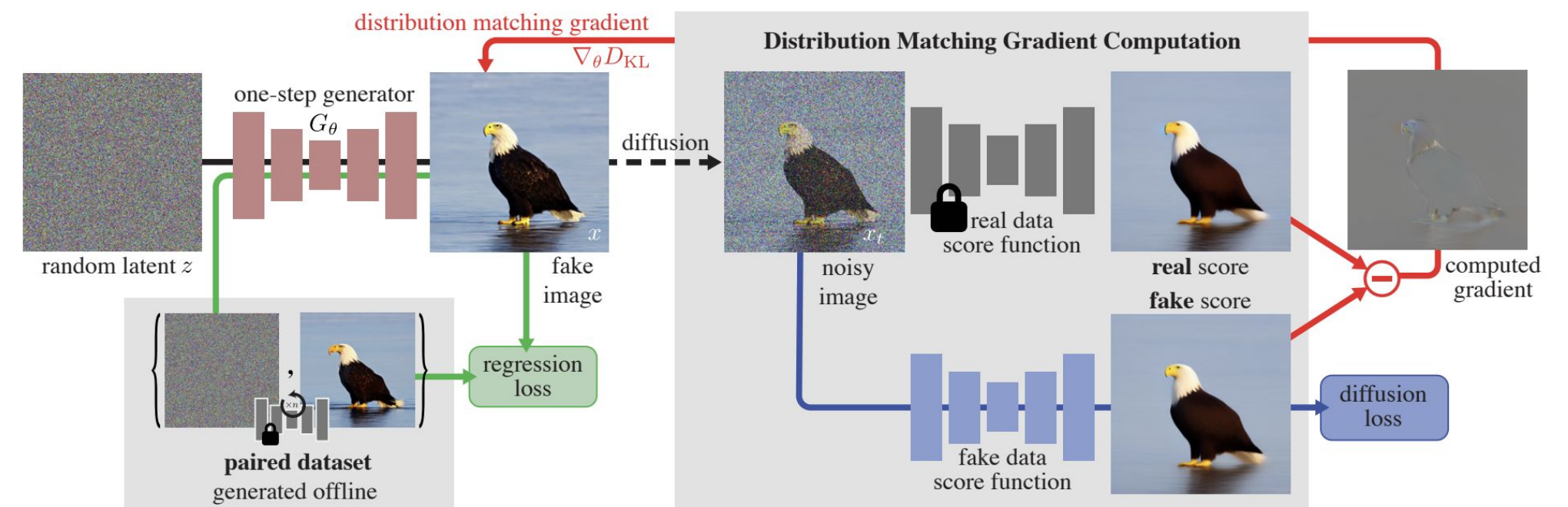
Supplementary

DMD :

regression-loss dependency $\rightarrow$ costly & limited

DMD2 :

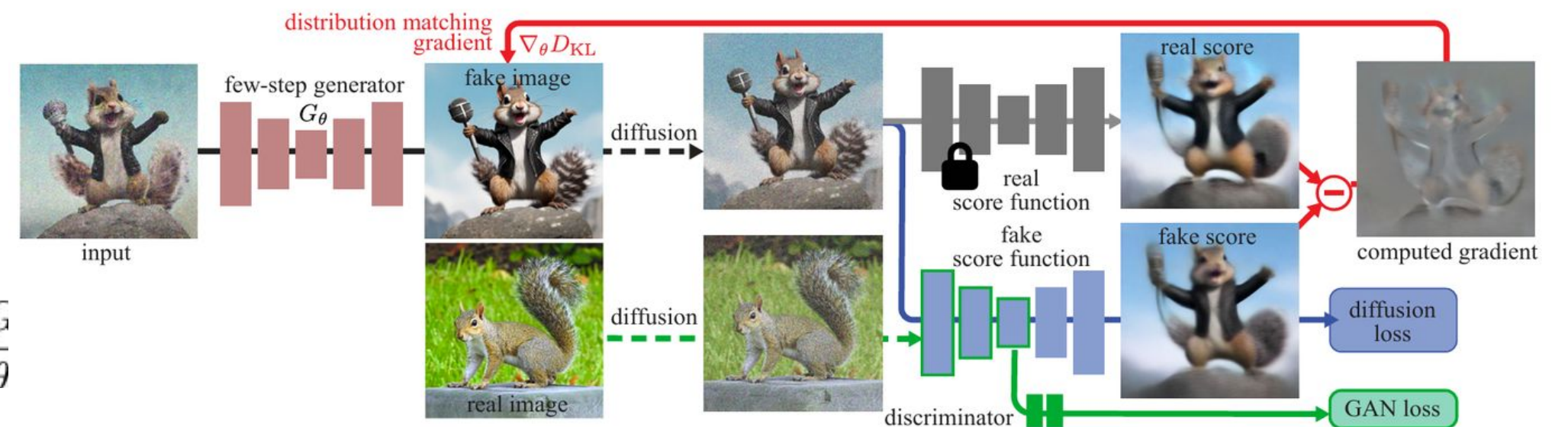
Large amount of data required



Distribution matching gradient

$$\nabla_{\theta} D_{KL} \simeq \mathbb{E}_{z, t, x, x_t} \left[w_t \alpha_t (s_{\text{fake}}(x_t, t) - s_{\text{real}}(x_t, t)) \frac{dG}{d\theta} \right]$$

$$w_t = \frac{\sigma_t^2}{\alpha_t} \frac{CS}{\|\mu_{\text{base}}(x_t, t) - x\|_1}$$



Subject-Driven Generation with DreamBooth

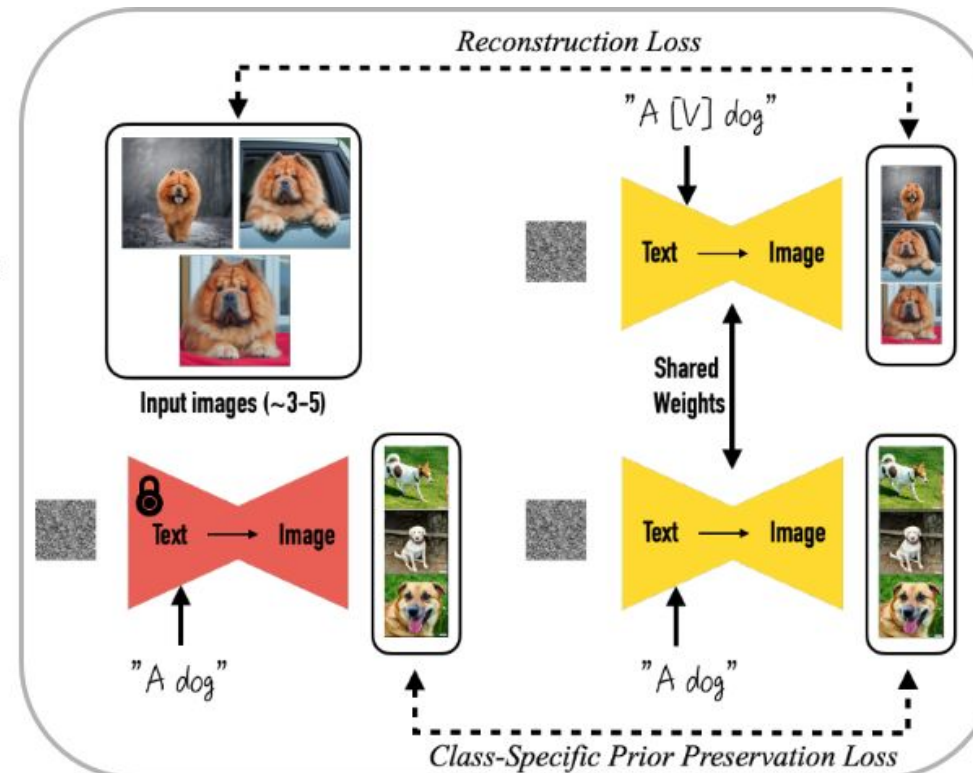
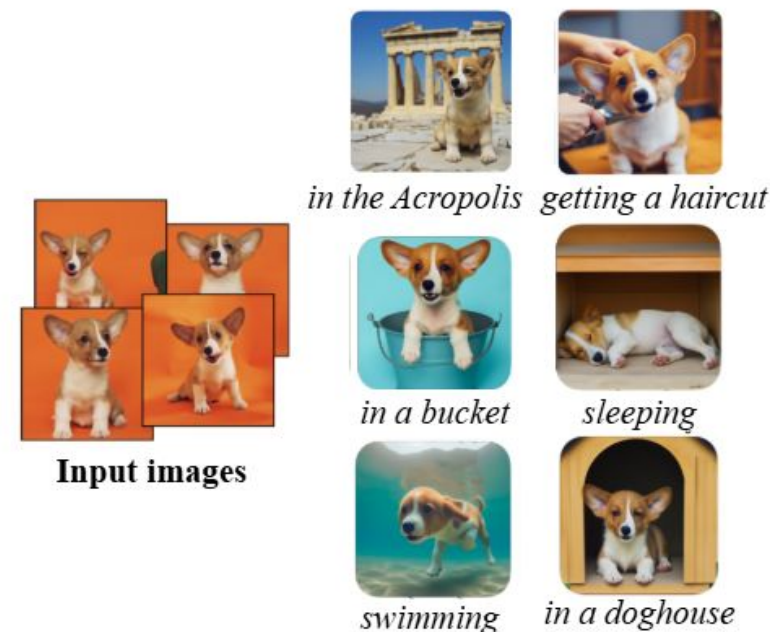
Supplementary

Standard Diffusion Loss

$$\mathbb{E}_{\mathbf{x}, \mathbf{c}, \epsilon, \epsilon', t} [w_t \|\hat{\mathbf{x}}_{\theta}(\alpha_t \mathbf{x} + \sigma_t \epsilon, \mathbf{c}) - \mathbf{x}\|_2^2]$$

Autogenous Class-Specific Prior Preservation Loss

$$+ \lambda w_{t'} \|\hat{\mathbf{x}}_{\theta}(\alpha_{t'} \mathbf{x}_{\text{pr}} + \sigma_{t'} \epsilon', \mathbf{c}_{\text{pr}}) - \mathbf{x}_{\text{pr}}\|_2^2]$$



- Subject binding via **unique token Identifier**
- **Few-shot fine-tuning**
- **Class Specific Prior Preservation Loss**

→ **Diverse personalized generation**

End.

